

Reliability Evaluation of Distribution Network Based on Sequential Monte Carlo Simulation

Qi Lu

Nari Technology Development
Limited Company
Nanjing, China
luqi@sgepri.sgcc.com.cn

Yujie Hua, Ye Shen

Nanjing University of
Science and Technology
Nanjing, China
627082016@qq.com

Qi Kang

New York Institute of
Technology
New York, America
winner908@163.com

Rong Zhang

Shanxi University of
Technology
Xi'an, China
zjf807807@sina.com

Abstract—In this paper, the reliability evaluation of distribution network is studied by using the sequential Monte Carlo simulation based on dual sampling. A reliability evaluation model of distribution network is established. The range influenced by fault is determined by the analysis method of node fault. And then, all kinds of reliability index are calculated. With matlab programming, IEEE RBTS Bus 6 test system is evaluated to prove that the method in this paper is effective. Then the simulation analysis is carried out for the cases that whether there is a standby transformer in power distribution network, whether the circuit breaker is reliable and whether there is a contact switch between the two feeders. The results show that the reliability of distribution network system can be enhanced by adding the standby transformer and the contact switch and improving the reliability of the circuit breaker.

Keywords—Distribution network; Reliability; Monte Carlo method; Sequential

I. INTRODUCTION

Distribution network reliability evaluation is a key part of the power network, but also the core of network and user connection. When the power distribution system has a fault and it is repaired or tested by power workers, the distribution network can't give users a sustained and effective power supply, which will affect people's life greatly [1-3]. So we need to put more work and material resources to study the reliability of the distribution system.

At present, there are two kinds of methods for reliability evaluation of distribution system, which mainly include analytical method and simulation method. The analytical method is widely used, but it can't be used to establish models for complex distribution network. Simulation method is also known as Monte Carlo simulation, it is more suitable for complex distribution network, and it can find the accidents that the other methods can't predict. It reflects the randomness of the load [1-3]. In this regard, the domestic and foreign scholars have done a lot of research. Among them, the distribution network is evaluated by considering the capacity constraint and power flow distribution in reference [4, 5]. By changing the parameters of the components, the factors affecting the reliability index are analyzed in reference [6]. Simulation analysis is carried out by using the method on whether there is isolation switch or branch line protection on the feeder to

minimum path in reference [7]. In this paper, a sequential Monte Carlo simulation based on dual sampling is used to evaluate the reliability of distribution network, and the reliability evaluation index of the system and load points are calculated. And the simulation analysis is carried out for the cases that whether there is a standby transformer in power distribution network, whether the circuit breaker is reliable and whether there is a contact switch between the two feeders. Finally, the measures to improve the reliability of the distribution network are presented.

II. SEQUENTIAL MONTE CARLO SIMULATION BASED ON DUAL SAMPLING

A. Sequential Monte Carlo Simulation

Sequential Monte Carlo simulation is a kind of simulation method, which considers the time sequence of the system, and can simulate various situations of the distribution network in normal operation. This method greatly reflects the randomness and uncertainty of the distribution network in the course of operation [8]. The main idea is showed as follows. Firstly, according to the failure rate and repair rate of each component, a random sample of each component is performed, then the random failure model and the repair model are formed. So the sequence of work and fault in the distribution network during the year can be determined. Secondly, the status of each moment of the distribution network is checked. If there is a component failure, the impact of each load and the whole system will be judged and the duration of each failure will be determined. After 8760 hours of the simulation, the reliability index of the load and the system can be obtained. Using the above method to simulate the distribution network for many years, all kinds of average index can be obtained according to the relevant equations. If the calculation results meet the requirements, the simulation is over [9].

B. Sampling Principle of Sequential Monte Carlo Simulation Based on Dual Sampling

Firstly, the state of each component is sampled. Assuming that the demand (x) is the expected value ($E(\xi)$) of a random variable (ξ). N random samplings are carried out according to

National Natural Science Foundation of China (61203129)

the probability distribution function of ξ . Its arithmetic mean value is $\bar{\xi} = \frac{1}{N} \sum_{i=1}^N \xi_i$. When the value of N is great, it can get:

$$P(\lim_{N \rightarrow \infty} \bar{\xi} = x) = 1 \quad (1)$$

According to the law of large numbers, we can prove that $\bar{\xi} = E(\xi) = x$. So $\bar{\xi}$ can be used as an estimated value of x .

For distribution network evaluation, the state of the each component of the power distribution system should be sampled, and each state has its own event probability ($P(x)$). In the sampling, the random number sequence generated by the computer can't be distributed evenly, so the dual sampling method is used to generate a negative correlation with each random number as the next sampling value. It makes the simulation results more accurate. Let $F(x)$ be a test of the state x , the expected values of the test results are as follows:

$$E(F) = \sum_{x \in X} F(x)P(x) \quad (2)$$

The expected value ($\hat{E}(F)$) of the test function (F) is obtained by formula (3):

$$\hat{E}(F) = \frac{1}{N_s} \sum_{i=1}^{N_s} F(X_i) \quad (3)$$

N_s represents the total number of sampling, x_i represents the status value of the i times, $F(x_i)$ is the test results of x_i . Because the $E(F)$ is an estimated value of $\hat{E}(F)$, the error is determined by the variance, which is shown in the formula (4):

$$V(\hat{E}(F)) = V(F) / N_s \quad (4)$$

$V(F)$ is the variance of the test function (F), and the estimated value is:

$$\hat{V}(F) = \frac{1}{N} \sum_{i=1}^{N_s} [F(x_i) - \hat{E}(F)]^2 \quad (5)$$

III. RELIABILITY EVALUATION OF DISTRIBUTION NETWORK BASED ON SEQUENTIAL MONTE CARLO SIMULATION

In this paper, the reliability evaluation model of distribution network based on sequential Monte Carlo simulation is established. When the state sample is carried out, the dual sampling method is also added to determine the component of the failure. In the search and judgment, the analysis method of node failure is used to analyze all the load points when the failure occurred.

A. Component Model

In this paper, the simplified model is used in the operation, and only the power distribution line and power distribution transformer will occur. The circuit breaker does not occur, but it is Possible to be failure to operate. Other components work normally. The time when the component works is the average duration of working hours (TTF). The time in which the com-

ponent is in a state of failure is called an average repair time (TTR). In the study process of this paper, the component's failure rate is a certain value, and the working curve of the component obeys exponential distribution. So TTF obeys the negative exponential distribution, and its probability density ($f(t)$) is:

$$f(t) = \begin{cases} \lambda e^{-\lambda t} & t \geq 0 \\ 0 & t < 0 \end{cases} \quad (6)$$

Thus, the failure distribution function of the component is:

$$F(t) = 1 - e^{-\lambda t} = \delta \quad (7)$$

In the formula, δ is a random number between (0, 1).

In combination with the above contents, the TTF of the component is calculated by the formula (8), and the TTR of the component is calculated by the formula (9). So the running state sequence of the components in the simulation process can be obtained.

$$TTF = -\ln \delta / \lambda \quad (8)$$

$$TTR = -\ln \delta / \mu \quad (9)$$

In the formula, λ is the failure rate of component, μ is the repair rate of component.

B. Failure Effect Analysis

When a component fails, in order to determine the affected load points and its failure time, the equivalent model of the network is established in this paper. The static component in the distribution network is represented by the node, and the connection between each two nodes is represented by the dynamic component. When a component fails in the system, the node failure analysis method is used to determine the affected load points and the failure duration in the whole system. The specific steps are as follows:

1) When the component i in the system fails, the position of the component is judged. If it is in the main feeder, go to step 2). If it is on the branch line, go to step 5).

2) If the fault component i is on the main feeder, search breaker or switch in the direction of the main power supply. When the circuit breaker or the switch is found, a random number (β) is generated and it is judged whether β is between 0 and p (p is a reliable operation rate). If β is between 0 and p , the search is stopped, and the switch works reliably. The first circuit breaker B1 and the first switch S1 are written down. If β is not between 0 and p , the search continues.

3) If there is no standby power in the system, the load point before the B1 is still working normally, and all the load points after B1 are in fault. If the load is between B1 and S1, the outage time is the operation time of S1. If the load is after S1, the fault time is the repair time of the component i .

4) If the system is equipped with standby power supply, the section switch S2 is searched against the direction of main power supply on the main feeder. As shown in 3). The loads before B1 still work normally, and the loads after B1 are in fault. If the load is between B1 and S1, the outage time is the

operation time of the sub switch. If it is between S1 and S2, the outage time is the repair time of the component i. If it is after S2, the fault time is the switching time of the contact switch.

5) Forward search the switch along the shortest path on the branch line. When a switch device is searched, a random number (β) is generated and it is judged whether it is between 0 and p. If β is between 0 and p, the search is stopped. The first circuit breaker B2 and the switch S3 are written down. The load point before B2 is still working normally, and all the load points after B2 break down. If β is not between 0 and p, the search continues. If the load point is between B2 and S3, the fault duration is the operation time of S3. And if it is after S3, the time of failure is the repair time of the component i.

C. Reliability Index

When the system is running for a period of time, the index of the load can be calculated by the following three formulas: the failure rate λ , the average outage duration r , the annual average outage duration U [10] :

$$\lambda_j = \frac{N_j}{\sum T_{uj}} \quad (10)$$

$$r_j = \frac{\sum T_{dj}}{N_j} \quad (11)$$

$$U_j = \frac{\sum T_{dj}}{\sum T_{dj} + \sum T_{uj}} \quad (12)$$

In the formula, λ_j represents the failure rate of the load points, T_u represents the normal working hours of the load points during the simulation, T_d represents the failure time of the load point during the simulation, N_j represents the total number of load failure.

The index of each load point can be obtained by using the above formulas. According to the calculation formulas of the system index, SAIDI, CAIDI, SAIFI and other indicators can be obtained.

D. Process of Reliability Evaluation Based on Sequential Monte Carlo Simulation

The specific process of the reliability evaluation of distribution network based on dual sampling Monte Carlo simulation is shown in Fig.1.

IV. EXAMPLES OF RELIABILITY EVALUATION OF DISTRIBUTION NETWORK

A. Reliability Evaluations of Different Structures

The selected example of this section is the main feeder F4 of IEEE RBTS Bus 6, The level of the feeder is 33kV, which includes 30 lines, 23 load points, 23 fuses, 23 power distribution transformers, 4 circuit breakers and 1 sub switch. The relevant data reference [11], the wiring diagram of the feeder is shown in Fig.2.

Case 1: The reliability of circuit breakers working reliably is 1, and the transformer has no reserve. Case 2: The reliability of circuit breaker working reliably is 0.8, and the transformer has no reserve. Case 3: The reliability of the circuit breaker working reliably is 1, and the transformer has a backup. The system is simulated by the method described in this paper. The simulation time is 20000 years. The reliability index of the three kinds of load points is shown in Table I, and the system reliability index is shown in Table II.

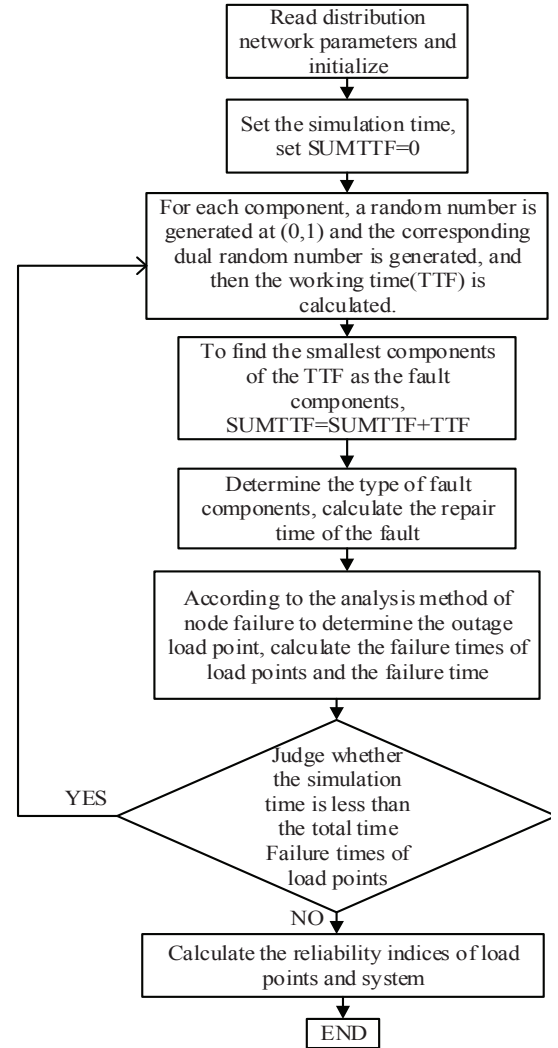


Fig. 1. Process of reliability evaluation of Monte Carlo simulation based on dual sampling

From the results of former two cases, it can be seen that the average failure rate of each load point will be increased when the circuit breaker is not reliable. In particular, the failure rate of the load point (LP1—LP10) which is not on the branch feeder is obviously increased, which leads to an increase in the annual average outage duration. Therefore, the reliability of the circuit breaker can affect the reliability index of load points and system.

For case 3, if the system is equipped with a spare transformer, the fault correction time can be shortened to 10 hours, which greatly reduces the annual average outage duration of the load point. Although the average power failure frequency of the system has not reduced, the users and the system have a certain degree of reduction on the average outage duration. It means that this measure can effectively improve the reliability of the system.

B. Reliability Evaluations of Different Systems

The example selected for this section is the feeder F1 and F2 of IEEE RBTS Bus 6. It consists of 26 lines, 13 load points, 13 fuses, 13 power distribution transformers, 2 circuit breakers, 13 sub switches and a contact switch. In the simulation, the load transfer is not considered. Assuming that the capacity of the transformer can satisfy all load work normally, it does not need to cut load to ensure the power supply quality of the system. The wiring diagram of the system is shown in Fig.3, the relevant data reference [11], and the simulation time is 20000 years. Case 4: there is a contact switch between two feeders. Case 5: there is no contact switch between two feeders. If there is no contact switch between F1 and F2, the reliability evaluation of the F1 is only conducted. The reliability index of the load point is shown in Table III.

As can be seen from Table III, after adding the contact switch, the total failure rate is not reduced. However, because of the failure time of the partial load points becomes the action time of the isolated switch, the average outage time and the average annual outage time are greatly reduced. So the power outage time of the load is reduced and the reliability of the distribution network is improved.

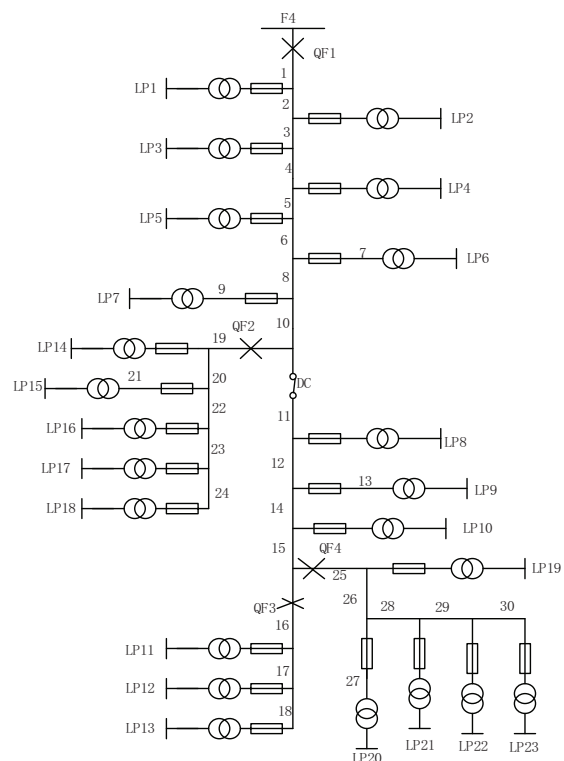


Fig. 2. Distribution system for IEEE RBTS BUS6 F4

TABLE I. RELIABILITY INDEX OF LOAD POINT

Load point	λ			U			r		
	Case 1	Case 2	Case 3	Case 1	Case 2	Case 3	Case 1	Case 2	Case 3
LP1	1.1840	1.4332	1.1844	8.5552	9.3021	5.7020	7.2256	6.4901	4.8140
LP2	1.1835	1.4326	1.1866	8.2833	8.9219	5.7232	6.9987	6.2274	4.8231
LP3	1.1861	1.4333	1.1849	8.4902	9.3149	5.7022	7.1580	6.4985	4.8121
LP4	1.1862	1.4335	1.1852	8.7245	9.4667	5.7345	7.3550	6.6035	4.8382
LP5	1.1853	1.4310	1.1864	8.5381	8.7634	5.7407	7.2030	6.1238	4.8386
LP6	1.2119	1.4581	1.2122	8.7682	8.8815	5.9513	7.2351	6.0908	4.9093
LP7	1.2202	1.4678	1.2215	8.7487	8.8845	6.0028	7.1696	6.0526	4.9141
LP8	1.1856	1.4328	1.1860	12.3136	14.4699	9.5697	10.3855	10.0984	8.0685
LP9	1.2090	1.4611	1.2113	12.3393	15.0769	9.7283	10.2057	10.3183	8.0313
LP10	1.1857	1.4331	1.1863	13.0164	14.8183	9.5707	10.9777	10.3398	8.0676
LP11	1.5789	1.7567	1.5770	15.5116	17.1148	12.7108	9.8240	9.7421	8.0597
LP12	1.5797	1.7551	1.5776	15.5476	16.7757	12.7243	9.8415	9.5581	8.0651
LP13	1.5796	1.7557	1.5780	15.2135	16.9977	12.7114	9.6306	9.6812	8.0549
LP14	1.7869	1.9270	1.7963	13.0711	13.2569	10.6591	7.3148	6.8792	5.9337
LP15	1.7975	1.9402	1.8112	13.3741	13.1529	10.9431	7.4402	6.7790	6.0416
LP16	1.7887	1.9256	1.7957	13.6633	12.5413	10.6244	7.6385	6.5126	5.9165
LP17	1.7869	1.9265	1.7970	13.1819	12.7779	10.6537	7.3766	6.6324	5.9285
LP18	1.7874	1.9279	1.7958	13.3519	13.4426	10.6331	7.4696	6.9725	5.9210
LP19	1.7843	1.9054	1.7910	17.4942	18.3657	14.4653	9.8043	9.6383	8.0763
LP20	1.8153	1.9408	1.8249	17.0095	19.0781	14.7413	9.3698	9.8298	8.0775
LP21	1.7839	1.9026	1.7927	17.4825	17.9624	14.4771	9.7996	9.4408	8.0752
LP22	1.7834	1.9049	1.7925	17.4953	18.5384	14.4770	9.8098	9.7315	8.0760
LP23	1.7834	1.9055	1.7905	17.3339	18.6272	14.4647	9.7191	9.7753	8.0784

TABLE II. RELIABILITY INDEX OF SYSTEM

index	SAIFI	SAIDI	CAIDI	ASAI	ENS
Case 1	1.4008	11.8464	8.4570	0.9986	61.9939
Case2	1.6076	12.5658	7.8167	0.9986	65.9898
Case3	1.4033	9.0067	6.4182	0.9990	48.6790

TABLE III. RELIABILITY INDEX OF LOAD POINT OF CASE 4 AND CASE 5

Load point	λ		r		U	
	Case4	Case5	Case4	Case5	Case4	Case5
LP1	0.3320	0.3366	11.8285	11.6433	3.9275	3.9191
LP2	0.3464	0.3483	11.4500	11.0976	3.9673	3.8662
LP3	0.3408	0.3453	11.0005	13.0231	3.7499	4.4973
LP4	0.3307	0.3727	11.8206	12.4231	3.9101	4.6305
LP5	0.3384	0.3426	11.2805	13.6083	3.8183	4.6633
LP6	0.3333	0.3319	11.9621	13.7840	3.9880	4.5761

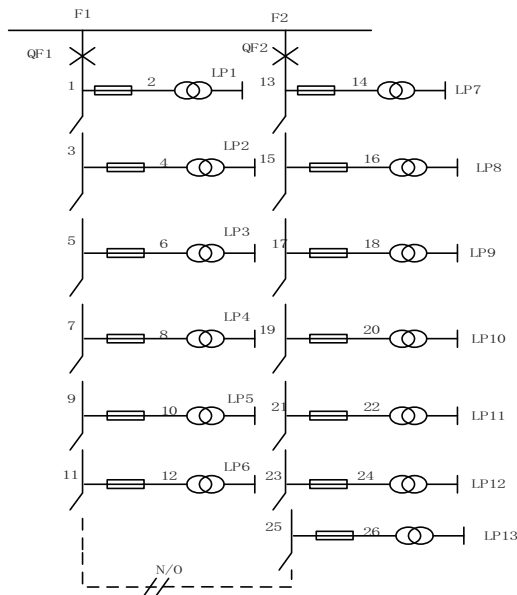


Fig. 3. Distribution system for IEEE RBTS BUS6 F1 and F2

V. CONCLUSION

In this paper, the sequential Monte Carlo simulation based on dual sampling is used to evaluate the distribution network, and the reliability index of the system is calculated. And the simulation analysis is carried out for the cases that whether there is a standby transformer in power distribution network, whether the circuit breaker is reliable and whether there is a contact switch between the two feeders. The results show that the reliability of the power distribution system can be enhanced by adding the standby transformer or contacting switch between two feeders or improving the reliability of the circuit breaker.

ACKNOWLEDGMENT

This work was supported by the National Natural Science Foundation of China (61203129).

REFERENCES

- [1] Guocheng Wan, Zhen Ren, and Xiang Tian, "Study on model of reliability-network-equivalent of distribution system reliability evaluation," Chinese Journal of Electrical Engineering, vol. 23, pp. 49-53, 2003.
- [2] Yuqin Xu, and Yingchao Wu, "Reliability evaluation for distribution system connected with wind-turbine generators," Grid Technology, vol. 35, pp. 154-158, 2011.
- [3] Can Chen, Wenchuan Wu, Boming Zhang, and Jianguang Qin, "Reliability evaluation for large-scale distribution network based on component groups," Grid Technology, vol. 36, pp. 81-86, 2012.
- [4] Li Shao, Kaigui Xie, Jin Wang, and Chun You, "Reliability evaluation algorithm of distribution network based on power flow estimation and section load shedding," Grid Technology, vol. 32, pp. 33-38, 2008.
- [5] Hua Zhao, Zhuding Wang, Kaigui Xie, and Wenyuan Li, "Comparative study on reliability assessment methods for medium voltage distribution network," Grid Technology, vol. 37, pp. 3295-3302, 2013.
- [6] Hui Zhang, Junhua Li, and Xiaoping Dong, "Arithmetic analysis of supply reliability based on monte carlo simulation," Electrical Technology, vol. 12, pp. 5-8, 2011.
- [7] Xiaojuan Zhang, Zerong Li, Zhongyuan Zhang, and Junbo Dong, "Distribution system reliability assessment based on minimal path method," Grid and Clean Energy, vol. 26, pp. 24-28, 2010.
- [8] Ming Ding, Jing Zhang, and Shenghu Li, "A sequential monte-carlo simulation based reliability evaluation model for distribution network," Grid Technology, vol. 28, pp. 38-42, 2004.
- [9] Huishi Liang, Lin Cheng, and Sige Liu, "Monte carlo simulation based reliability evaluation of distribution system containing microgrids," Grid Technology, vol. 35, pp. 76-81, 2011.
- [10] R. Billinton, and P. Wang, "Teaching distribution system reliability evaluation using Monte Carlo simulation," IEEE Trans on Power Systems, vol. 11, pp. 1670-1676, 1996.
- [11] R. Billinton, and S. Jonnavithula, "A test system for teaching overall power system reliability assessment," IEEE Transactions on Power Delivery, vol. 11, pp. 1670-1676, 1996.